Report on
"**The Quality of Lattice Sequences for Arbitrary Sample Size**"
by L. Matiukha, Y. Ding and F.J. Hickernell

As mentioned in the final sentences of Section 2, bottom of page 3, the aim of this manuscript is to extend a result from Hickernell and Niederreiter (2003) on an upper bound for some quality measure. Minimal numerical illustrations are added. I admit that I did not verify all mathematical details in the proofs.

I can't agree with the way Section 2 – Background starts. Historically (the authors use the word!) lattices and lattice rules were introduced in a much more general way than suggested in the beginning of this paragraph. Simply look into the referenced book by Sloan & Joe (1984). What is described in this section is a rank 1 rule. These receive almost all attention in recent literature in the first place because they are much easier to analyse and reduce search spaces. One should not forget that.

Almost all remarks below are minor but there are many...

page 2, last lines: It's a pity that nothing more is included about these 'generalized integers' that seem crucial for selecting the generator $\mathbf{h}$. It's a pity that it remains unclear in the sequel for which powers of base b the point set is a full lattice. (A remark that popped up in my mind again when reading the intro of section 3.)

Somehow the presentation on page 3 should make it clear that the top half, up to equation (2), is for one dimension. The sentence that starts with "Hickernell and Niederreiter" is about a multidimensional setting.

page 3, Fig. 1: The fonts used in Figure 1 are different from those in the text, and the color might be selected more distinguishable. Please include $\mathbf{h}$ and Δ in the caption.

What caught my eyes first in Section 3 is that the authors use $\sqrt{-1}$ to denote the imaginary unit instead of the more common i. I can understand why –because i is already used in the cubature sum– but strictly speaking this is not correct. The imaginary unit is defined by $\mathrm{i}^2 + 1 = 0$. I believe it's up to the editors to decide whether or not this is acceptable, taking into account consistency of notations within a journal.

page 4, line 7: Please add a sentence to motivate the use of Banach spaces; see also further remark.

page 4, line 13: insert 'an' to obtain 'using an arbitrary point set'

page 4, equation 4: observe the argument of $\hat{I}_n$. The dot should be an $\mathbf{x}$, and the closing bracket is positioned too low. Note that this typo and mispositioning appears also on many other places further in this manuscript, and I will not repeat it.

page 4, just below equation (4) is mentioned that Holder's inequality is used, but the is no inequality in the derivation above... The last equality should be an inequality.

page 4, last but one line: I suggest to write 'takes on a simple form shown below'.

page 5, equation (5): the definition of B should be moved upwards as it already appears earlier.

page 5, last line -8: is it really 'does not exist' or is it 'not known'?

page 5, last line -7: 'there exists...' maybe add that this is only an existence result

page 6, first sentence of section 4: insert 'for' to obtain 'result for arbitrary n'

page 7, line 1: use log (math mode) instead of *log*

pages 6-7, Theorem 1: I wonder if the comments on C_P at the end of a.i and b are best placed inside the Theorem or not ...

page 7, beginning of section 4.1: How do you know, and can the reader see, that the upper bound is 'reasonably conservative'?

page 8, middle: I don't see the relation between a 'good choice of **h**' and the derivation in the same sentence (14). I don't have a clue what a good choice means – see above on the missing info on **h**.

page 10, last line -1: when mentioning coordinate weights, please refer back to the Banach space setting on page 3.

page 12, beginning of section 4.5: I suggest to include the used vector **h** here instead of referring to the 'default', which might be time dependent. And why should I believe that it satisfies the conditions of the derivation in this manuscript ('generalised integer' on page 2 and the 'there exists' condition in the theorems)? It follows from the details but I suggest to mention that this is a 2-dimensional example. Assuming you ran a similar experiment in more dimensions, can a comment on that being added?

page 13: 'exploring' the performance with just one extra integral is of course very limited but I understand that the authors want to illustrate something. So please make clear that this example fits in the space for which the theory is derived.

page 14: be consistent in the captions of the figures. Fig. 3 has 'six-dimensional' and Fig. 4 '6-dimensional' it its description.

page 15, line 8: The stronger norm gives the largest number?

page 15, line 16: the 'tight bound' is an equality...

page 17–18, section 5.3 and figures: please make **h** explicit. Make it clear in the text that Fig. 6 is an example in 6 dimensions and Figs. 7-8 only in 2 dimensions. Surprisingly you used less points in the lower-dimensional

illustration. Now this section ends with the observation. Please add a sentence with your conclusion from this. End with sharing the insight you obtained from this experiment.

page 17, last sentence before Figure: 'weighs' → 'weights'

page 18: 'the smoothness of the integrand is limited' is easy to misinterpret in my opinion.

page 18, Discussion: 'early termination' is, both in the beginning and here, used as motivation for the arbitrary n. Now that I reached this part of the manuscript that feels a bit uneasy. The part on page 16 about optimal weights depends on having the point set in advance, I believe...